

FIELD HANDBOOK

DETAILED GUIDE FOR OPERATION, MAINTENANCE & TECHNICAL REFERENCE

// Insulator Cutting Machine



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// DISCLAIMER NOTICE.



Insulator Cutting Machine

This system is an early-stage prototype of a custom-built in-house solution developed entirely using a combination of original design, open-source references, third-party components, and AI-assisted engineering. It is a fully functional working base version currently undergoing active development. By using this system, you acknowledge and agree to the following:

1. HARDWARE FOUNDATION & DEVELOPMENT APPROACH:

This system is built using a combination of custom-designed wiring, PCB layouts, and selected third-party components. Where designs reference publicly available schematics, open-source hardware, or third-party modules (including those developed under AIMS-EMS or AIMS-NET), appropriate credit is given to their respective sources. No original claims of ownership are made over foundational electronic concepts or industry-standard circuit designs.

2. FIRST REVISION – OPERATIONAL CAUTION:

This is an initial release version and should be treated as an evolving system. It is subject to hardware revisions, wiring changes, and component substitutions over time. Although operational, users may encounter unexpected behavior, component incompatibilities, or performance inconsistencies. Independent validation of all outputs and safety checks is strongly advised before use in critical or production applications.

3. REFERENCE-ASSISTED DESIGN PROCESS:

System design has been assisted by AI tools, online references, and third-party technical resources for schematic generation, component selection, and optimization. These tools accelerate development but do not guarantee compliance with all applicable standards. Human validation, physical testing, and safety checks remain mandatory before deployment.

4. USER FEEDBACK & TECHNICAL ISSUE REPORTING:

Users are strongly encouraged to report any hardware faults, wiring anomalies, component failures, or unexpected system behavior directly to the Kaizen Team. Suggestions for design improvements, component upgrades, or usability enhancements are welcome and may be incorporated into future revisions.

5. CONFIDENTIALITY & INTELLECTUAL PROPERTY RIGHTS:

All schematics, PCB designs, wiring diagrams, custom configurations, and related hardware assets developed under this system are the intellectual property of its developer, SUNCHIRIN INDUSTRIES MALAYSIA. Unauthorized sharing, copying, reproduction, or distribution of any design files or technical documentation — especially to external parties, clients, or customers — is strictly prohibited without prior written consent from authorized personnel. This system is intended for internal use only.

By operating, modifying, or distributing this system, you agree to the terms outlined in this disclaimer.

Your feedback and involvement in shaping this system into a robust & reliable industrial tool are invaluable.

Thank you.

// PLC PROGRAM COMMENTS.



Insulator Cutting Machine

INPUT (X):

- X0 = LEN = LENGTH INPUT TRIGGER
- X1 = START = START PROCESS
- X2 = STOP = STOP PROCESS
- X3 = EMER = EMERGENCY STOP
- X4 = SAFE = SAFETY SENSOR SWITCH (TRIGGER)
- X5 = OPEN = OPEN CENTER PLATE
- X6 = CLOSE = CLOSE CENTER PLATE
- X7 = CUTL = HOLD MANUAL CUT LEFT BUTTON
- X10 = CUTR = HOLD MANUAL CUT RIGHT BUTTON

OUTPUT (Y):

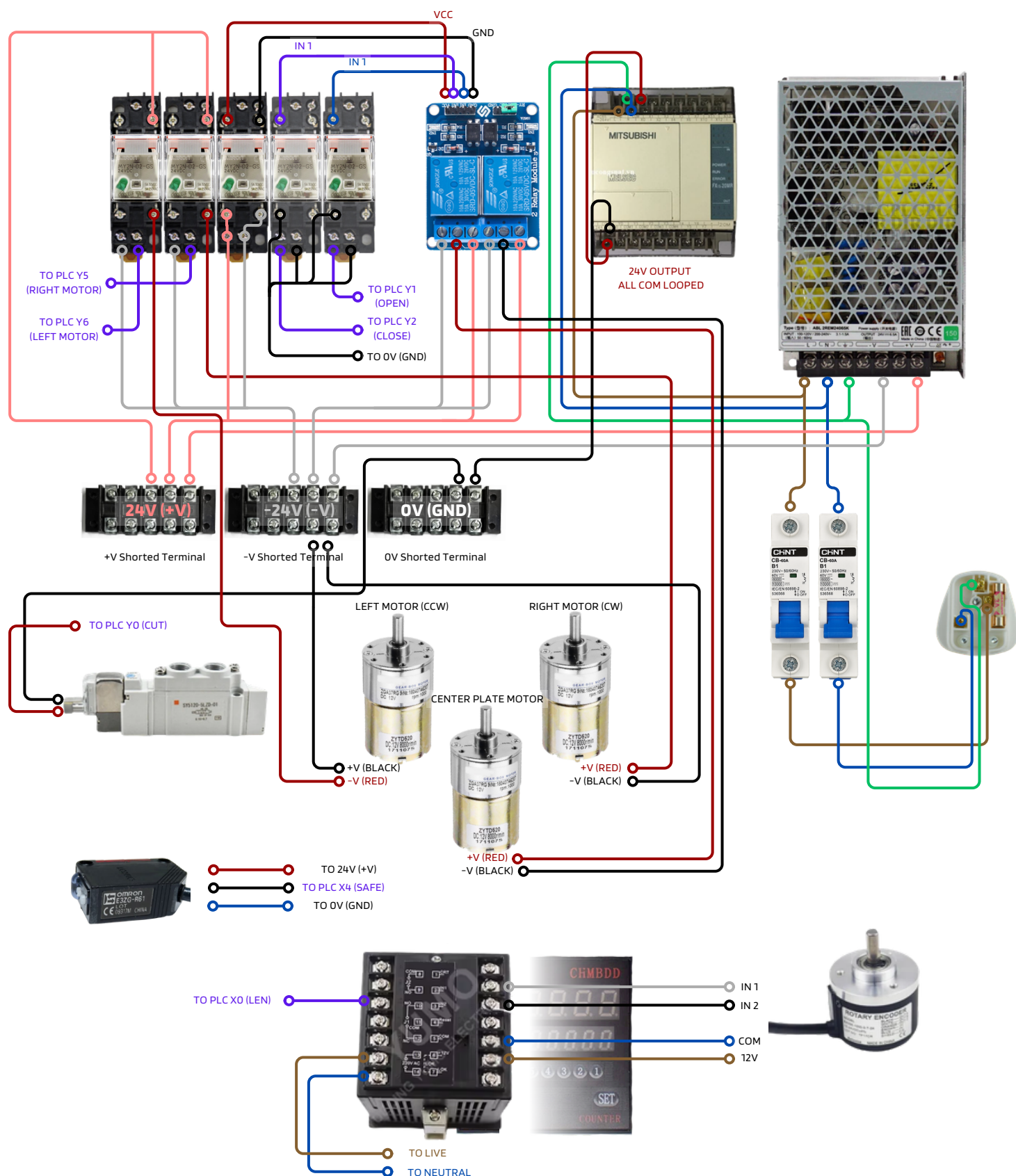
- Y0 = CUT = CUT INSULATOR
- Y1 = OPEN = OPEN CENTER PLATE
- Y2 = CLOSE = CLOSE CENTER PLATE
- Y3 = RBLT = RIGHT BELT (RIGHT MOTOR CW RUN)
- Y4 = LBLT = LEFT BELT (LEFT MOTOR CCW RUN)
- Y5 = LEDG = GREEN LED
- Y6 = LEDY = YELLOW LED
- Y7 = LEDR = RED LED + BUZZER

MEMORY (M):

- M0 = STARTOP = START OPERATION (AUTO)
- M1 = MOVEBELT = MOVE BELT (LEFT & RIGHT)
- M2 = CUTOP = CUTTING OPERATION
- M3 = STARTBTN = START BUTTON (HMI)
- M4 = STOPBTN = STOP BUTTON (HMI)
- M5 = RESETBTN = RESET COUNTER BUTTON (HMI)
- M6 = JOGBTN = MANUAL JOG CONVEYOR BELT BUTTON (HMI)
- M7 = OPENBTN = OPEN PLATE BUTTON (HMI)
- M8 = CLOSEBTN = CLOSE PLATE BUTTON (HMI)

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// WIRING DIAGRAM.
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// COMPONENT DETAILS.



MITSUBISHI MELSEC PLC

- FX1S-20MR
- 1 Unit



SCHNEIDER ELECTRIC PSU

- ABL2REM24065K
- 1 Unit
- Output 24V 6.5A
- 150W Supply



SONGLE RELAY SWITCH

- SRD-05VDC-SL-C
- 1 Unit
- W/ Jumper from VCC to JD-VCC
- 2 Channels



PLUG POINT HEAD

- UK TYPE
- 1 Unit
- With 13A fuse



CHNT MCB

- CB-60A
- 2 Units
- Rating for 240VAC 50/60Hz incoming



DIGITAL COUNTER

- ATK72 CHMBDD
- 1 Unit



OMRON RELAY SOCKET BASE

- PYF08A-E
- 5 Units
- 8 PINS



SMC SOLENOID VALVE CONTROL

- SY5120-5LZD-01
- 1 Unit
- 2 Silencers



OMRON RELAY SWITCHES

- MY2N-GS
- 5 Units
- 24VDC



PHOTOELECTRIC SENSOR

- OMRON E3ZG-R61
- 1 Unit
- Set as Pulse UP
- PNP-Sourcing



DC GEAR BOX MOTOR

- ZHENG ZGA37RG80 GEARHEAD
- CENTER SHAFT - 6mm Diameter
- 3 Unit
- ZHENG ZYTD520 DC MOTOR
- DC 24V 5000r/min 60RPM 3Nm

